MEASURABLE DENSITY BOUNDS HOLD FOR ALL COLOURINGS OF THE PLANE WITH ALGEBRAIC FORBIDDEN DISTANCES

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Abstract. OpenAI recently proved that every colouring of the plane with five colours gives the same colour to two points at distance 1. The proof transfers an arbitrary proper colouring of the plane to a *weak measurable* one, and then excludes weak measurable 5-colourings. We observe that the transfer works, with the same proof, for every set of algebraic forbidden distances and for fractional colourings. Upper bounds for the density of measurable sets avoiding given distances therefore become lower bounds for arbitrary colourings; all these lower bounds rest on OpenAI's rigidity theorem, which OpenAI formalized in Lean; we formalized the transfer for proper colourings (not its fractional version) on top of their formalization. Consequences: the fractional chromatic number of the plane equals $1/m_1(\mathbb{R}^2)$, where $m_1(\mathbb{R}^2)$ is the largest density of a measurable planar set with no two points at distance 1, so it lies between 4.0485 and 4.36; and the least independence ratio of an n-point unit-distance graph tends to $m_1(\mathbb{R}^2)$, as conjectured by Dúcz and Varga. If the distances 1 and 2 are both forbidden, at least 7 colours are needed, improving the bound 6 of Exoo and Ismailescu; the same holds for $\{1, \sqrt{3}\}$, and at least 8, 9, 10 and 12 colours are needed for $\{1, 11/6\}$, $\{1, 40\}$, $\{1, 100\}$ and $\{1, 1000\}$. Uniformly in d, $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$ for every real algebraic $d \geq 6.52$, and the bound rises step by step to 13 for every real algebraic $d \geq 54\,300$; from $d = 12.3$ on, each step is proved by a single certificate for a whole half-line of d. The distances 1, 100 and 10^4 force 27 colours, 1, 10, 100 and 1000 force 26, and 1, 30, 900 and 27000 force 48. For suitable sets of k integer distances more than 4.0485^k colours are needed, in the plane and in every higher dimension, so the chromatic number with k forbidden distances grows exponentially in k, whereas Erdős expected polynomial growth. Davies has just proved the analogous statements in dimension at least 4 by a different method. Using OpenAI's theorem itself at a small scale and density bounds at a large one, we show that the distances 1 and N force at least 25 colours for every large algebraic N, although 19.01 colours suffice fractionally, and, with de Grey's graph, at least 18 colours for every algebraic $N \geq 10^{25}$; iterating, k distances can force at least $6 \cdot 4.0486^{k-1}$ colours. The density bounds we use are linear-programming bounds of Oliveira Filho and Vallentin, which we certify in interval arithmetic, and theorems of Bukh and of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki.

1. Results

For a set D of positive real numbers, $\chi(\mathbb{R}^2, D)$ is the least number of colours needed to colour the plane so that no two points whose distance lies in D get the same colour. The Hadwiger–Nelson problem is the case $D = \{1\}$; OpenAI [22] proved $\chi(\mathbb{R}^2) \geq 6$, and $\chi(\mathbb{R}^2) \leq 7$ is Isbell's hexagonal colouring [27]. For two distances $\{1, d\}$ the best known lower bound was 6: Owings, Tetiva and Huddleston [23] showed $\chi(\mathbb{R}^2, \{1, (1+\sqrt{5})/2\}) \geq 6$ (as reported in [4, 10]), Exoo and Ismailescu [13] proved $\chi(\mathbb{R}^2, \{1, 2\}) \geq 6$, Parts [24] gave a small 6-chromatic two-distance graph, and De Neve, Vanden Kerchove, Colle, Tavernier and Pickavet [10] gave 6-chromatic graphs for $d = \sqrt{3}$ and $d = (\sqrt{6} + \sqrt{2})/2$, two cases claimed earlier in the Polymath16 project [25]; Chybowska-Sokół, Junosza-Szaniawski and Węsek [4] record as open whether some pair $\{1, d\}$ forces 7 colours.

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the observation, wrote the programs, ran the checks and helped to draft this note. Four separate AI agents refereed the note with their own programs. The hard input is OpenAI's transfer theorem [22]. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

OpenAI's proof has two steps. Their Theorem 1.3 (the transfer) says that the plane has a proper k-colouring if and only if it has a *weak measurable* k-colouring: a measurable colouring in which the pairs of points at distance 1 with the same colour have measure zero. Their Theorem 1.4 excludes weak measurable 5-colourings. The transfer rests on a rigidity theorem for invariant measures on the character group of the algebraic plane [22, Thm. 2.3]. Its proof uses neither that the forbidden distance is 1 nor that a colouring uses finitely many colours rather than being fractional. We record the two generalisations (Theorems 11 and 12) and their consequences.

Write $m_D(\mathbb{R}^2)$ for the supremum of the upper densities of measurable sets $A \subseteq \mathbb{R}^2$ with no two points at a distance in D, and $m_1(\mathbb{R}^2) = m_{\{1\}}(\mathbb{R}^2)$. Croft's construction gives $m_1(\mathbb{R}^2) \geq 0.229364$ [6], and Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1] proved $m_1(\mathbb{R}^2) \leq 0.2470$, confirming a conjecture of Erdős that $m_1(\mathbb{R}^2) < 1/4$. For the fractional chromatic number we use two versions, defined in Section 2: the finitary one $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$, the supremum over finite graphs drawn in the plane with edge lengths in D, and the fractional chromatic number $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ of the whole infinite graph.

Theorem 1. *Let D be a finite set of positive real algebraic numbers. Then*

$$\chi(\mathbb{R}^2, D) \;\geq\; \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \;=\; \chi_{\mathrm{f}}(\mathbb{R}^2, D) \;=\; \frac{1}{m_D(\mathbb{R}^2)}.$$

The lower bound $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$, and with it every lower bound below, rests on OpenAI's rigidity theorem [22, Thm. 2.3], which OpenAI formalized in Lean; the upper bound $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/m_D(\mathbb{R}^2)$ is elementary.

One distance.

Corollary 2. $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, *and* $4.0485 < 1/0.2470 \leq \chi_{\mathrm{f}}(\mathbb{R}^2) \leq 1/0.229364 < 4.36$.

The constants in [1, §8] give $m_1(\mathbb{R}^2) < 0.2469975$, hence $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0486$. Matolcsi, Ruzsa, Varga and Zsámboki [19] proved $\chi_{\mathrm{f}}(\mathbb{R}^2) \geq 4$ and conjectured that $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 4$ and that $\chi_{\mathrm{f}}(G) < 4$ for every finite unit-distance graph G [19, Conj. 1]; the value 4 had also been conjectured in [1, Remark after Cor. 2]. Dúcz and Varga [12] disproved this conjecture: they found a unit-distance graph G_{29} whose geometric fractional chromatic number exceeds 4.0007, which gives a finite unit-distance graph with independence ratio below $1/4$ and $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0007$. Corollary 2 gives a second disproof and determines $\chi_{\mathrm{f}}(\mathbb{R}^2)$ in terms of $m_1(\mathbb{R}^2)$. It also answers the question of [19] whether $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2)$, and with [19, Thm. 2] it shows that the finitary independence ratio of the plane equals the measurable one, which [19] remark would fail if their conjecture held. Let $f(n)$ be the least independence number of the unit-distance graph of a set of n points in the plane, the function of Erdős's question [16] answered by Dúcz and Varga.

Corollary 3. $f(n) \geq m_1(\mathbb{R}^2)\, n$ *for every n, and* $f(n)/n \to m_1(\mathbb{R}^2)$ *as* $n \to \infty$. *In particular* $f(n) < 0.2470\, n$ *for all large n.*

The first statement is the classical averaging bound $m_1(\mathbb{R}^2) \leq \alpha_1(\mathbb{R}^2)$ [1, Remark after Cor. 2], [19, (14)], [12, Claim 1]; the new content is the limit, that is, $\alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$. Dúcz and Varga conjectured that $f(n)/n = m_1(\mathbb{R}^2) + o(1)$, and in particular that $m_1(\mathbb{R}^2)$ is the density $0.2293647\ldots$ of Croft's construction [12, Conj. 1]. Corollary 3 proves the first statement; the second remains open.

Two distances.

Corollary 4.

$$\chi(\mathbb{R}^2, \{1, 2\}) \geq 7, \qquad \chi(\mathbb{R}^2, \{1, \sqrt{3}\}) \geq 7, \qquad \chi(\mathbb{R}^2, \{1, (\sqrt{6}+\sqrt{2})/2\}) \geq 7,$$
$$\chi(\mathbb{R}^2, \{1, 11/6\}) \geq 8, \qquad \chi(\mathbb{R}^2, \{1, 40\}) \geq 9,$$
$$\chi(\mathbb{R}^2, \{1, 100\}) \geq 10, \qquad \chi(\mathbb{R}^2, \{1, 1000\}) \geq 12.$$

Moreover $\chi(\mathbb{R}^2, \{1, d\}) \geq 17$ *for every sufficiently large real algebraic number d.*

The finite bounds come from linear programs certified in Section 6, and the last statement from Bukh's theorem [2]; products of hexagonal colourings give $\chi(\mathbb{R}^2, \{1, d\}) \le 49$. For *measurable* colourings, bounds of this kind were known: Oliveira Filho and Vallentin [21, §4] report, from their linear program, the bound $0.141577 < 1/7$ for the density of a measurable set avoiding the distances 1 and $j_{1,2}/j_{1,1} \approx 1.8309$, a ratio of zeros of J_1, and, combining Bukh's theorem with their bound for $m_1(\mathbb{R}^2)$, they show $\inf_{d_1,d_2} m_{\{d_1,d_2\}}(\mathbb{R}^2) \le m_1(\mathbb{R}^2)^2 \le 0.072046$; so pairs of distances whose density is close to $m_1(\mathbb{R}^2)^2$ need 14 colours for measurable colourings (17 with the bound of [1]). What is new here is that such bounds hold for arbitrary colourings; the bounds for $\{1, 2\}$ and $\{1, \sqrt{3}\}$, which use clique inequalities, also seem to be new for measurable colourings.

The same linear program gives bounds for whole ranges of d at once.

Theorem 5. *Let $d \ge 1$ be a real algebraic number and $7 \le k \le 13$. Then $\chi(\mathbb{R}^2, \{1, d\}) \ge k$ whenever $d \ge d_k$, where*

$$d_7 = 6.52, \quad d_8 = 12.05, \quad d_9 = 24.3, \quad d_{10} = 52.9, \quad d_{11} = 140.2, \quad d_{12} = 601, \quad d_{13} = 54\,300.$$

Moreover $\chi(\mathbb{R}^2, \{1, d\}) \ge 7$ for $1.75 \le d \le 1.93$, and $\chi(\mathbb{R}^2, \{1, d\}) \ge 8$ for $1.83 \le d \le 1.855$. The same bounds hold for $\chi(\mathbb{R}^2, \{1, 1/d\}) = \chi(\mathbb{R}^2, \{1, d\})$, and, for measurable colourings, for every real d.

Theorem 5 contains the finite bounds of Corollary 4 except those for 2, $\sqrt{3}$ and $(\sqrt{6} + \sqrt{2})/2$; see Figure 1. Scaling the plane by $1/d$ turns $\{1, d\}$ into $\{1/d, 1\}$, which gives the equality with $1/d$. For measurable colourings neither the transfer nor algebraicity is needed: Proposition 17 and Lemma 18 do not use them, and one class of a measurable proper k-colouring avoiding $\{1, d\}$ is a measurable set avoiding $\{1, d\}$ of upper density at least $1/k$. For measurable colourings bounds of this size were known for one pair [21, §4] and, by Bukh's theorem, for all d beyond an unspecified threshold (see the discussion after Corollary 4); there the new content is the explicit thresholds and ranges. For each k a single function serves every $d \ge D_k$ (Lemma 18 and Table 3; $D_k = d_k$ except $D_8 = 12.3$), and the two bounded ranges and $[12.05, 12.3]$ are covered by finitely many functions, each serving an interval of d (Section 7). The thresholds are close to the best this linear program can give: in floating point its value is below $1/(k-1)$ from about $d = 6.503$, 12.03, 23.87, 52.85, 139.9 and 599.2 on, for $k = 7, \dots, 12$, and for $k = 13$ from somewhere between 50 000 and 54 300. For $1 < d < 6.5$ it is below $1/6$ only for d between about 1.747 and 1.934, and on $(2, 6.5)$ it is largest near $d = 2.59$, where it is $1/4.68$. As $d \to \infty$ its value tends to $\mu_0^2/(1 + \mu_0)^2 = 1/12.13\ldots$, where $\mu_0 = -\min J_0 = 0.40275\ldots$: the square of its value $\mu_0/(1 + \mu_0) = 0.287119$ for one distance [21, §3.2] (Remark after Lemma 18). So for two distances it gives at most 13 colours (for $d \ge 20$ by that Remark, and in floating point for every d); Theorem 8 gives more for very large d.

Many distances.

Corollary 6. *For every $k \ge 1$ there is a set D of k positive integers with $\chi(\mathbb{R}^d, D) \ge \chi(\mathbb{R}^2, D) \ge m_1(\mathbb{R}^2)^{-k} > 4.0485^k$ for every $d \ge 2$. Moreover*

$$\sup\{\chi_{\mathrm{f}}(\mathbb{R}^2, D) : D \subseteq F \cap (0, \infty),\ |D| = k\} = m_1(\mathbb{R}^2)^{-k},$$

where F is the field of real algebraic numbers.

Erdős [15] defined $L_2(r)$ as the largest chromatic number of the plane with r forbidden distances and asked: "Does $L_2(r)$ increase polynomially or exponentially? I expect that $L_2(r) < r^c$ for some absolute constant c." The question is Problem #706 on Bloom's list [14], where it is stated for finite point sets, which gives the same $L_2(r)$ by the de Bruijn–Erdős theorem; Croft, Falconer and Guy ask whether the growth is exponential [5, Problem G11] (as quoted by Bukh [2]), and the best lower bound in the plane was of order $k\sqrt{\log k}$ (see [2, 20]). By Corollary 6 and the product colouring,

$$4.0485^r < L_2(r) \le 7^r,$$

so $L_2(r)$ grows exponentially, contrary to Erdős's expectation. Bukh [2] proved exponential growth for measurable colourings and remarked that it is "very likely that the usual chromatic

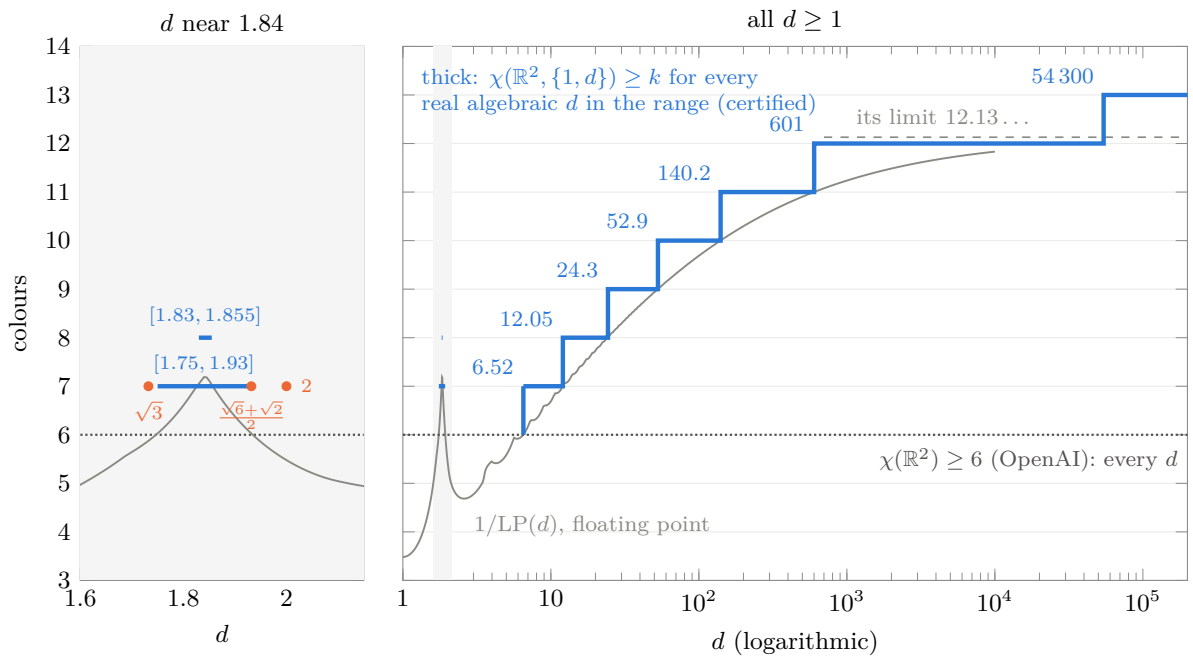


FIGURE 1. Lower bounds for $\chi(\mathbb{R}^2, \{1, d\})$, $d \geq 1$; the same hold for $1/d$. Thick blue: Theorem 5, for every real algebraic d in the range. Orange: the pairs $\{1, \sqrt{3}\}$, $\{1, 2\}$ and $\{1, (\sqrt{6}+\sqrt{2})/2\}$ of Corollary 4, certified with clique inequalities or at a single d. Grey: $1/\mathrm{LP}(d)$, where $\mathrm{LP}(d)$ is the value of the linear program in floating point (not a proof); a certified bound below $1/(k-1)$ gives k colours, so the blue steps are less than one colour above the grey curve, and at most slightly below it. Dashed: the limit $12.13\ldots$ of the grey curve. Dotted: $\chi(\mathbb{R}^2) \geq 6$ [22], which holds for every d. Left: the shaded range $1.6 \leq d \leq 2.15$ on a linear scale.

number does not share this behavior"; Corollary 6 shows that it does, for integer distances. Davies [7] has just proved exponential growth for arbitrary colourings of $\mathbb{R}^d$, $d \geq 4$, with a different transfer (see the Remarks); Corollary 6 gives it in every dimension $d \geq 2$ (for $d \geq 4$ Davies's base $1/m_1(\mathbb{R}^d)$ is larger, for distances with rational squares). Like all our lower bounds, it rests on [22, Thm. 2.3]. Bukh also conjectured that $\chi(\mathbb{R}^d, D)$ is bounded independently of D when the elements of D are algebraically independent [2]; Theorem 11 says nothing about such D.

The same linear programs give explicit sets (Table 2):

Corollary 7. *$\chi(\mathbb{R}^2, \{1, 40, 1600\}) \geq 22$, $\chi(\mathbb{R}^2, \{1, 10, 100, 1000\}) \geq 26$, $\chi(\mathbb{R}^2, \{1, 100, 10^4\}) \geq 27$ and $\chi(\mathbb{R}^2, \{1, 30, 900, 27000\}) \geq 48$.*

These numbers are far below $m_1(\mathbb{R}^2)^{-k}$, which is approached only by very spread distances, but the sets are explicit and the bounds are certified.

Two scales. The lower bounds so far come from densities alone, so for two distances they never exceed the fractional chromatic number, which is below 19.01 for every pair $\{1, N\}$ (Corollary 9). Using OpenAI's theorem itself at one scale and densities at the other gives more.

Theorem 8. *Let D_1 and D_2 be finite sets of positive real algebraic numbers, let c be a positive integer with $c \leq \chi(\mathbb{R}^2, D_1)$, and let W be a finite graph drawn in the plane with all edge lengths in D_2. There is N_0 such that*

$$\chi(\mathbb{R}^2, D_1 \cup N D_2) \;\geq\; \chi_c(W) \;\geq\; c\,\chi_f(W)$$

for every real algebraic number $N \geq N_0$. Consequently, with $c = \chi(\mathbb{R}^2, D_1)$,

$$\chi(\mathbb{R}^2, D_1 \cup N D_2) \;\geq\; \chi_c(\mathbb{R}^2, D_2) \;\geq\; \frac{c}{m_{D_2}(\mathbb{R}^2)}$$

for all large real algebraic N.

Here a *c-fold colouring* of a graph with k colours gives every vertex a set of c of the k colours, with disjoint sets at adjacent vertices, and the c-fold chromatic number χ_c is the least such k (Stahl's c-th chromatic number; see [18] for the plane). So $\chi_1 = \chi$, $\chi_c \geq c\,\chi_{\mathrm{f}}$ for finite graphs, and $\chi_c(\mathbb{R}^2, D) = \chi_c(G(\mathbb{C}, D))$.

At the scale of D_1, every disc of a fixed radius meets at least c classes of the measurable colouring given by Theorem 11, each in a set of measure bounded below, because a finite graph that needs c colours fits in the disc; at the scale of ND_2 these classes behave like a c-fold colouring. For graphs, the analogue is Geller and Stahl's formula $\chi(G[H]) = \chi_{\chi(H)}(G)$ for the lexicographic product [17], and Theorem 8 is the lower half of a geometric version; whether $\chi(\mathbb{R}^2, D_1 \cup ND_2) \to \chi_{\chi(\mathbb{R}^2, D_1)}(\mathbb{R}^2, D_2)$ as $N \to \infty$ is open (products of colourings give only $\chi(\mathbb{R}^2, D_1)\,\chi(\mathbb{R}^2, D_2)$). The proof is in Section 5. For $c = 6$ no explicit finite graph is known, so N_0 is not explicit; with explicit inputs it is (the remark on explicit pairs in Section 5).

Corollary 9. (1) $\chi(\mathbb{R}^2, \{1, N\}) \geq 25$ *for every sufficiently large real algebraic number N, whereas* $\chi_{\mathrm{f}}(\mathbb{R}^2, \{1, N\}) \leq m_1(\mathbb{R}^2)^{-2} < 19.01$ *for every $N > 0$.*

(2) *There are integers* $1 < d_2 < d_3 < d_4$ *with* $\chi(\mathbb{R}^2, \{1, d_2\}) \geq 25$, $\chi(\mathbb{R}^2, \{1, d_2, d_3\}) \geq 102$ *and* $\chi(\mathbb{R}^2, \{1, d_2, d_3, d_4\}) \geq 413$, *and for every $r \geq 1$ there is a set D of r positive integers with* $\chi(\mathbb{R}^d, D) \geq \chi(\mathbb{R}^2, D) \geq 6 \cdot 4.0486^{r-1}$ *for every $d \geq 2$. Hence* $L_2(r) > 1.48 \cdot 4.0486^r$.

So for two distances the chromatic number can exceed the fractional chromatic number by a factor above $25/19.01 > 1.3$. For a single distance, OpenAI's theorem and Corollary 2 give the factor $6/4.36 > 1.37$. The bound $6/m_1(\mathbb{R}^2)$ that Theorem 8 gives for $\{1, N\}$ lies between 24.29 and 26.16, while products of hexagonal colourings give $\chi(\mathbb{R}^2, \{1, N\}) \leq 49$. The bound 25 needs only some finite unit-distance graph W with $\chi_{\mathrm{f}}(W) > 4$, such as the graph of Dúcz and Varga [12], so it does not depend on Theorem 12 or on [1]; the constants 102 and 413 do (from $\chi_{\mathrm{f}}(W) > 4.0007$ alone one gets 25, 101 and 405).

None of these results comes with an explicit finite graph: as in [22], the finite graphs exist by compactness.

2. Notation

Identify the plane with $\mathbb{C}$. Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$, let $E = F(i)$ be the plane of algebraic points and $K = \{u \in E : |u| = 1\}$ its group of rotations, as in [22, Def. 2.1]. For $D \subseteq (0, \infty)$ and $X \subseteq \mathbb{C}$ let $G(X, D)$ be the graph on X in which two points are adjacent when their distance lies in D. A set of real algebraic numbers is countable, and for $d \in F$ the points of E at distance d from 0 are exactly those of dK.

Fractional chromatic numbers. For a graph G with countably many vertices, $\chi_{\mathrm{f}}(G)$ is the least t such that some Borel probability measure on the closed set of independent sets of G in $\{0,1\}^{V(G)}$ gives every vertex probability at least $1/t$ of lying in the random set. For finite G this is the usual fractional chromatic number, and by weak-$*$ compactness $\chi_{\mathrm{f}}(G)$ is the supremum over the finite subgraphs of G. We write $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$ for the supremum of $\chi_{\mathrm{f}}(H)$ over finite graphs H drawn in the plane with all edge lengths in D, the *finitary* fractional chromatic number of [19] when $D = \{1\}$, and $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ for the infimum of $\sum_I w(I)$ over finitely supported weightings $w \geq 0$ of independent sets of $G(\mathbb{C}, D)$ with $\sum_{I \ni x} w(I) \geq 1$ for every point x. Clearly $\chi_{\mathrm{f},0} \leq \chi_{\mathrm{f}}$. Since F is real closed, Tarski's transfer principle lets every finite graph drawn with edge lengths in $D \subseteq F$ be redrawn with vertices in E; so, for $D \subseteq F$,

$$\chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D)) \quad \text{and} \quad \chi(\mathbb{R}^2, D) = \chi(G(E, D)).$$

Densities. The upper density of a measurable $A \subseteq \mathbb{C}$ is

$$\overline{\delta}(A) = \limsup_{R \to \infty} \frac{|A \cap B(0, R)|}{|B(0, R)|}.$$

A set *avoids* D if no two of its points are at a distance in D, and $m_D(\mathbb{R}^2)$ is the supremum of $\overline{\delta}(A)$ over measurable sets A avoiding D. For finite D it is also the supremum of the densities of periodic sets avoiding D (Lemma 13; the same proof works with squares in place of discs), so it

does not matter whether densities are taken over discs, as here, or over squares, as in [21, 2]. A measurable set A *weakly avoids* D if

$$\int_{B(0,R)} \int_{S^1} 1_A(x)\, 1_A(x+du)\, d\sigma(u)\, dx = 0 \qquad \text{for all } d \in D,\ R > 0,$$

where σ is the normalized arc length. A *weak measurable k-colouring avoiding D* is a measurable map $c : \mathbb{C} \to \{1, \dots, k\}$ whose classes weakly avoid D.

3. The transfers

From [22] we use Lemma 4.1 (the continuous vectors of a measure-preserving action of E form L^2 of a factor, and the translations extend to a strongly continuous action T_x, $x \in \mathbb{R}^2$, on it), Theorem 2.3 (every K-invariant Borel probability measure on the character group of the discrete group E that gives measure zero to the continuous characters is Haar measure), the measurable sampling step in the proof of Proposition 4.2, and Lemma 4.3. Lemma 4.1 and Theorem 2.3 involve no distance; Lemma 4.3 and the sampling step concern the unit circle, and we redo them at radius d (Lemma 10 and the proofs below).

Lemma 10. *If a measurable set A weakly avoids D, then the set A^* of points of density one of A avoids D, and $|A \setminus A^*| = 0$.*

Proof. This is [22, Lemma 4.3] at radius d. Let x, $y = x + v \in A^*$ with $|v| = d \in D$. For small $h > 0$ choose x' uniformly in $B(x, h)$ and, independently, $v' = du'$ with u' uniform on the arc $\{u \in S^1 : |du - v| < h\}$. Then $P(x' \notin A) \to 0$. The point $x' + v'$ is a mixture of uniform points of discs of radius h, so it has a density at most $1/(\pi h^2)$, and it lies in $B(y, 2h)$; hence $P(x' + v' \notin A) \le |B(y, 2h) \setminus A|/(\pi h^2) \to 0$. For small h both points lie in A with positive probability, while the law of (x', u') is absolutely continuous with respect to $dx\, d\sigma(u)$. This contradicts weak avoidance. The second statement is Lebesgue's density theorem. □

Theorem 11. *Let $D \subseteq F$ be a set of positive real algebraic numbers and $k \ge 1$. If $G(E, D)$ has a proper k-colouring, the plane has a weak measurable k-colouring avoiding D; conversely, such a colouring gives a proper k-colouring of $G(\mathbb{C}, D)$. In particular $\chi(\mathbb{R}^2, D) \ge 1/m_D(\mathbb{R}^2)$.*

Proof. This is the proof of [22, Thm. 1.3] with the unit circle replaced by the circles of radii $d \in D$.

(1) Let $\Omega_D \subseteq \{1, \dots, k\}^E$ be the closed set of proper colourings of $G(E, D)$, nonempty by hypothesis. Translations by E and rotations by K preserve it. The group $E \rtimes K$ is amenable, so averages of a point mass along a Følner sequence have an invariant weak-$*$ limit μ.

(2) Let f_i be the indicator that the origin has colour i. Properness gives $\langle f_i, T_z f_i \rangle = 0$ for $z \in dK$, $d \in D$.

(3) By [22, Lemma 4.1], $f_i = p_i + b_i$ where p_i is the conditional expectation on the factor of continuous vectors; $0 \le p_i \le 1$, $\sum_i p_i = 1$, both parts are rotation-fixed, and the spectral measure of b_i is K-invariant and null on the continuous characters. By [22, Thm. 2.3] it is a multiple of Haar measure, so $\langle b_i, T_z b_i \rangle = 0$ for $z \in E \setminus \{0\}$; the cross terms vanish because the continuous subspace and its orthogonal complement are translation invariant. Hence $\langle p_i, T_z p_i \rangle = 0$ for $z \in dK$.

(4) The T_x act strongly continuously and dK is dense in the circle of radius d, so $\langle p_i, T_x p_i \rangle = 0$ whenever $|x| \in D$.

(5) The sampling step of [22, Prop. 4.2] gives jointly measurable fields $H_i(x, \omega)$ representing $T_x p_i$. Since D is countable, a single sample ω makes all the integrals

$$\int_{B(0,m)} \int_{S^1} H_i(x, \omega) H_i(x + du, \omega)\, d\sigma(u)\, dx \qquad (m \in \mathbb{N},\ d \in D,\ \text{all } i)$$

vanish and has $\sum_i H_i(x, \omega) = 1$ for almost every x. Giving x the least i with $H_i(x, \omega) > 0$ defines the weak measurable colouring.

(6) Conversely, by Lemma 10 the density-one points of the classes form disjoint sets avoiding D whose union is conull. Every finite configuration has a translate inside that union, so every finite subgraph of $G(\mathbb{C}, D)$ is k-colourable, and the de Bruijn–Erdős theorem [8] gives a proper colouring. One of the k sets has upper density at least $1/k$, so $k \geq 1/m_D(\mathbb{R}^2)$. □

Formal verification. Theorem 11, in the form used in Section 5, is formally verified in Lean 4 (version 4.34.1, Mathlib commit `d13f23b7`) on top of OpenAI's formalization of their proof (`https://github.com/openai/math`, commit `fd4aeeb2`): for every set D of positive real algebraic numbers, a proper k-colouring of $G(\mathbb{C}, D)$ gives a measurable k-colouring whose classes weakly avoid every $d \in D$ (`OAI.Ext.proper_to_borel_weak_distances`), and k pairwise disjoint measurable sets, each avoiding D, whose union is conull (`OAI.Ext.proper_to_measurable_classes`); Lemma 10 at radius r is `OAI.Ext.weak_density_proper_radius`. The proofs use only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`); the files are in `lean/external/openai-five/distances/` of [26]. Not formalized: the converse of Theorem 11, the inequality $\chi \geq 1/m_D$, Theorem 12, the lemmas of Section 5 and the density bounds.

Theorem 12. *For every set $D \subseteq F$ of positive real algebraic numbers, $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$.*

Proof. Let $t \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D))$ and let P be a Borel probability measure on the compact metrizable space Ω of independent sets of $G(E, D)$ with $P(z \in I) \geq 1/t$ for all $z \in E$. Averaging the push-forwards of P under $E \rtimes K$ along a Følner sequence and taking a weak-$*$ limit gives an invariant probability μ with $\mu(0 \in I) \geq 1/t$, since each push-forward has this property and the event is clopen. Let $f = 1[0 \in I]$. It is rotation-fixed, and $\langle f, T_z f\rangle = \mu(0 \in I,\ z \in I) = 0$ for $z \in dK$, $d \in D$, because I is independent. Steps 3 and 4 of the proof of Theorem 11 apply to the single function f, with $\sum_i p_i = 1$ (which is used only in Step 5) replaced by $\mathbb{E}_\mu p = \mathbb{E}_\mu f$: for $p = \mathbb{E}(f \mid \text{continuous factor})$ we get $0 \leq p \leq 1$, $\mathbb{E}_\mu p \geq 1/t$ and $\langle p, T_x p\rangle = 0$ whenever $|x| \in D$.

Take a jointly measurable $H : \mathbb{C} \times \Omega \to [0, 1]$ with $H(x, \cdot) = T_x p$ in $L^2(\mu)$ for almost every x (the sampling step of [22, Prop. 4.2]). For $m \in \mathbb{N}$ and $d \in D$,

$$\int_\Omega \int_{B(0,m)} \int_{S^1} H(x,\omega) H(x+du,\omega)\, d\sigma(u)\, dx\, d\mu(\omega) = \int_{B(0,m)} \int_{S^1} \langle T_x p, T_{x+du} p\rangle\, d\sigma(u)\, dx = 0,$$

so, D being countable, for μ-almost every ω all these inner integrals vanish. Let $X_n(\omega) = |B(0,n)|^{-1} \int_{B(0,n)} H(x,\omega)\, dx \in [0, 1]$. Since $\mathbb{E}_\mu H(x, \cdot) = \langle T_x p, 1\rangle = \langle p, 1\rangle \geq 1/t$ for almost every x ($T_x 1 = 1$ and T_x is unitary), $\mathbb{E}_\mu X_n \geq 1/t$, and the reverse Fatou lemma gives $\mathbb{E}_\mu \limsup_n X_n \geq 1/t$. Hence $\limsup_n X_n \geq 1/t$ on a set of positive measure, and we choose ω in it and in the conull set above. Let $A = \{x : H(x,\omega) > 0\}$. Then $1_A \geq H(\cdot,\omega)$, so $\overline{\delta}(A) \geq 1/t$; and $H(x,\omega)H(x+du,\omega) = 0$ for almost every (x, u) gives $1_A(x) 1_A(x+du) = 0$ almost everywhere, so A weakly avoids D. By Lemma 10, A^* avoids D and $\overline{\delta}(A^*) \geq 1/t$. Hence $m_D(\mathbb{R}^2) \geq 1/t$. □

4. The upper bound and the corollaries

Lemma 13. *Let D be finite, $M = \max D$ and $\varepsilon > 0$. There are $\eta > 0$ and a periodic set A' with period lattice $N\mathbb{Z}^2$, each period a finite union of squares, that avoids $D_\eta = D + [-\eta, \eta]$ and has density at least $m_D(\mathbb{R}^2) - \varepsilon$.*

Proof. Let A be measurable, avoiding D, with $\overline{\delta}(A) > m_D - \varepsilon/4$, and fix $L > 0$. For the radii R realising the upper density, with R large, the squares of the grid $L\mathbb{Z}^2$ meeting $B(0, R)$ (at most $\pi(R + \sqrt{2}L)^2/L^2$ of them) have average density of A above $m_D - \varepsilon/2$, so one of them, S, has $|A \cap S| \geq (m_D - \varepsilon/2)L^2$. By inner regularity there is a compact $C \subseteq A \cap S$ with $|C| \geq (m_D - 3\varepsilon/4)L^2$. Its distance set $\{|x - y| : x, y \in C\}$ is compact and misses the finite set D, so it misses $D_{2\eta}$ for some $\eta \in (0, 1)$. Let U be the union of the closed squares of a grid of squares of diameter less than $\eta/2$ that meet C. It lies in the $\eta/2$-neighbourhood of C, so it avoids D_η, and $|U| \geq |C|$. The set $A' = U + N\mathbb{Z}^2$ with $N = L + M + 2$ avoids D_η, since points of different copies are more than $M + 1$ apart, and has density $|U|/N^2 \geq m_D - \varepsilon$ once L is large. □

Proof of Theorem 1. By Theorem 12, $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D$, and $\chi \geq \chi_{\mathrm{f},0}$ always. Let A' be as in Lemma 13, with density $\delta \geq m_D - \varepsilon$ and fundamental square $Q = [0, N)^2$. For $n \geq 1$ take the n^2 translates $A' + s$, $s \in (N/n)\mathbb{Z}^2 \cap Q$; each avoids D. A point x lies in $A' + s$ if and only if $x - s \in A'$, so the number of these translates containing x is the number of points of the grid $x - (N/n)\mathbb{Z}^2$ in one period of A'. One period is a finite union of squares, so its boundary meets $O(n)$ cells of that grid, uniformly in x, and the count is at least $\delta n^2 - O(n)$ for every x. Giving each translate the weight $1/(\delta n^2 - O(n))$ is a fractional colouring of $G(\mathbb{C}, D)$ with total weight tending to $1/\delta$. Hence $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/(m_D - \varepsilon)$ for every $\varepsilon > 0$. □

Proof of Corollaries 2–6. Corollary 2 is Theorem 1 with $D = \{1\}$, together with $0.229364 \leq m_1 \leq 0.2470$ [6, 1]; [1, §8] proves $m_1 \leq (w_T + \nu)/(1 - \mu)$ with $w_T = 0.24697262945998308$, $\nu = 10^{-5}$ and $\mu = 6 \cdot 10^{-5}$, which is less than 0.2469975.

Corollary 3. If X is a set of n points, its unit-distance graph G has $\chi_{\mathrm{f}}(G) \leq \chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, so $\alpha(G) \geq n/\chi_{\mathrm{f}}(G) \geq m_1(\mathbb{R}^2)\, n$. Conversely, [19, Thm. 2] states $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/\alpha_1(\mathbb{R}^2)$, where $\alpha_1(\mathbb{R}^2)$ is the infimum of $\alpha(G)/|V(G)|$ over the unit-distance graphs G of finite point sets in the plane, so $\inf_n f(n)/n = \alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$ by Corollary 2. Placing two point sets far apart shows $f(a + b) \leq f(a) + f(b)$, so by Fekete's lemma $f(n)/n$ converges to $\inf_n f(n)/n = m_1(\mathbb{R}^2)$.

Corollary 4. The finite bounds follow from Theorem 11 and the density bounds of Table 1. For the last statement, Bukh's Theorem 1 [2] gives $m_{\{1,d\}}(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^2 \leq 0.2470^2 < 1/16$ as $d \to \infty$; Theorem 11 then gives $\chi(\mathbb{R}^2, \{1, d\}) \geq 17$ for large algebraic d. Corollary 7 follows in the same way from Table 2.

Corollary 6. Bukh's Theorem 1 gives $m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^k$ for $D = \{1, t, \ldots, t^{k-1}\}$ as $t \to \infty$. For integers t, Theorem 11 gives $\chi(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$; since $\chi(\mathbb{R}^2, D)$ is an integer and $1/m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^{-k}$, we get $\chi(\mathbb{R}^2, D) \geq m_1(\mathbb{R}^2)^{-k}$ for large t. Restricting a colouring of $\mathbb{R}^d$ to a plane gives $\chi(\mathbb{R}^d, D) \geq \chi(\mathbb{R}^2, D)$. For the supremum, Theorem 1 gives $\chi_{\mathrm{f}}(\mathbb{R}^2, D) = 1/m_D(\mathbb{R}^2)$ for finite $D \subseteq F$; Bukh proves $m_D(\mathbb{R}^2) \geq m_1(\mathbb{R}^2)^k$ whenever $|D| = k$ [2], and the sets $\{1, t, \ldots, t^{k-1}\}$ show that the bound $m_1(\mathbb{R}^2)^{-k}$ is approached. (By [21, Thm. 5.1], $m_D(\mathbb{R}^2) \leq 2^{-k}$ as soon as consecutive ratios exceed some $r(k)$ that their proof makes effective, which gives $\chi(\mathbb{R}^2, D) \geq 2^k$ for such D made of algebraic numbers.) □

5. Two scales

In this section $B(x, R)$ is the open disc, σ_d is the normalized arc length on the circle of radius d about 0, and $(f * \sigma_d)(x) = \int_{S^1} f(x + du)\, d\sigma(u)$. For a bounded measurable $f \geq 0$ and $d > 0$ let

$$P_d(f) = \limsup_{L \to \infty} \frac{1}{|B(0, L)|} \int_{B(0,L)} f\,(f * \sigma_d)\, dx,$$

the upper density of pairs at distance d, and $P_d(S) = P_d(1_S)$; a set that avoids d has $P_d(S) = 0$.

Lemma 14. *Let G be a finite graph drawn in the plane with all edge lengths in D_1 and $\chi(G) \geq c$, with n vertices, all in $B(0, R_0)$. Let $A_1, \ldots, A_k$ be measurable sets, each avoiding D_1, whose union is conull. Then $k \geq c$, and for every $x \in \mathbb{C}$ at least c indices i satisfy*

$$|A_i \cap B(x, 2R_0)| \;\geq\; \frac{|B(x, 2R_0)|}{4nk}.$$

Proof. Let $\Delta = B(x, 2R_0)$, let J be a set of $\min(c - 1, k)$ indices i with the largest values of $|A_i \cap \Delta|$, and let $U = \Delta \setminus \bigcup_{j \in J} A_j$. For $t \in B(x, R_0)$ the vertices of $G + t$ lie in Δ. If none of them were in U, each would lie in some A_j with $j \in J$; choosing one such j for every vertex would give a proper colouring of G with at most $c - 1$ colours, since two adjacent vertices are at a distance in D_1 and no A_j contains two such points. Hence every $t \in B(x, R_0)$ lies in $\bigcup_{v \in V(G)} (U - v)$, and $|U| \geq |B(0, R_0)|/n = |\Delta|/(4n) > 0$. The union of the sets A_i is conull, so $|U| \leq \sum_{i \notin J} |A_i \cap \Delta|$. Therefore some index lies outside J, which forces $k \geq c$ and $|J| = c - 1$, and the largest of the $k - c + 1$ terms of this sum is at least $|\Delta|/(4n(k - c + 1)) \geq |\Delta|/(4nk)$. By the choice of J, the $c - 1$ indices in J satisfy the same bound. □

Lemma 15. *Let A be a measurable set avoiding the distance $d>0$, let $R>0$ and $\rho(x)=|A\cap B(x,R)|/(\pi R^2)$. For every $L>0$,*

$$\int_{B(0,L)}\rho\,(\rho*\sigma_d)\,dx \;\le\; \sqrt{\frac{R}{\pi d}}\,|A\cap B(0,L+d+R)|, \qquad \text{so} \qquad P_d(\rho)\le\sqrt{\frac{R}{\pi d}}\,\overline{\delta}(A).$$

Proof. Let h be the indicator of $A\cap B(0,L+d+R)$, $\varphi=1_{B(0,R)}/(\pi R^2)$ and $F=h*\varphi$. For $|x|<L$ and $|u|=1$ the discs $B(x,R)$ and $B(x+du,R)$ lie in $B(0,L+d+R)$, so $F(x)=\rho(x)$ and $F(x+du)=\rho(x+du)$. Since $F\ge 0$, Plancherel's theorem gives

$$\int_{B(0,L)}\rho\,(\rho*\sigma_d)\,dx \;\le\; \int_{\mathbb{R}^2}F\,(F*\sigma_d)\,dx \;=\; \int_{\mathbb{R}^2}|\hat h(\xi)|^2\,\Lambda(2\pi R|\xi|)^2\,J_0(2\pi d|\xi|)\,d\xi,$$

where $\hat f(\xi)=\int f(x)e^{-2\pi i\langle x,\xi\rangle}dx$, $\hat\sigma_d(\xi)=J_0(2\pi d|\xi|)$, $\hat\varphi(\xi)=\Lambda(2\pi R|\xi|)$ and $\Lambda(s)=2J_1(s)/s$. As A avoids d, $h(x)h(x+du)=0$ for all x and u, so $0=\int h\,(h*\sigma_d)\,dx=\int|\hat h(\xi)|^2J_0(2\pi d|\xi|)\,d\xi$, and the right-hand side equals $\int|\hat h(\xi)|^2\,(\Lambda(s)^2-1)J_0(qs)\,d\xi$ with $s=2\pi R|\xi|$ and $q=d/R$.

We bound $(1-\Lambda(s)^2)\,|J_0(qs)|$. First, $|\Lambda|\le 1$ because $\hat\varphi$ is the Fourier transform of a probability density. For $0\le s\le 2\sqrt2$ the series $\Lambda(s)=\sum_{m\ge0}(-1)^m(s/2)^{2m}/(m!\,(m+1)!)$ alternates with nonincreasing terms, so $\Lambda(s)\ge 1-s^2/8\ge 0$ and $1-\Lambda(s)^2\le 2(1-\Lambda(s))\le s^2/4$; for $s>2\sqrt2$, $1-\Lambda(s)^2\le 1<s^2/4$. Second, $|J_0(y)|\le\min\bigl(1,\sqrt{2/(\pi y)}\bigr)$ for $y>0$ (Section 6). For $s\le 2$ the product is at most $\frac{s^2}{4}\sqrt{2/(\pi qs)}=\frac14 s^{3/2}\sqrt{2/(\pi q)}\le(\pi q)^{-1/2}$, and for $s\ge2$ at most $\sqrt{2/(\pi qs)}\le(\pi q)^{-1/2}$. Hence the integral is at most $(\pi q)^{-1/2}\int|\hat h|^2=\sqrt{R/(\pi d)}\,|A\cap B(0,L+d+R)|$. Dividing by $|B(0,L)|$ and letting $L\to\infty$ gives the second inequality. □

Lemma 16. *Let W be a finite graph drawn in the plane, let $c,k\ge1$ be integers with $k<\chi_c(W)$, and let $S_1,\dots,S_k$ be measurable sets such that every point of the plane lies in at least c of them. Then for every $N>0$*

$$\sum_{i=1}^{k}\sum_{ww'\in E(W)}P_{N|w-w'|}(S_i)\;\ge\;1.$$

Proof. For $x\in\mathbb{C}$ and $\theta\in S^1$ let $T(w)=\{i: x+N\theta w\in S_i\}$, a set of at least c indices. If $T(w)\cap T(w')=\emptyset$ for every edge ww', then c-element subsets of the sets $T(w)$ form a c-fold colouring of W with $k<\chi_c(W)$ colours, which is impossible. Hence

$$\sum_i\sum_{ww'\in E(W)}1_{S_i}(x+N\theta w)\,1_{S_i}(x+N\theta w')\;\ge\;1$$

for all x and θ. Integrate over $x\in B(0,L)$ and over θ (normalized Haar measure on S^1), and let $r=\max_w|w|$. For an edge ww', the substitution $y=x+N\theta w$ and the fact that $\theta(w'-w)$ is uniformly distributed on the circle of radius $|w'-w|$ give

$$\int_{S^1}\int_{B(0,L)}1_{S_i}(x+N\theta w)\,1_{S_i}(x+N\theta w')\,dx\,d\theta\;\le\;\int_{B(0,L+Nr)}1_{S_i}\,(1_{S_i}*\sigma_{N|w-w'|})\,dy.$$

Divide by $|B(0,L)|$ and take $\limsup_{L\to\infty}$; since $|B(0,L+Nr)|/|B(0,L)|\to1$ and the limit superior of a finite sum is at most the sum of the limits superior, the claim follows. □

Proof of Theorem 8. The graph $G(\mathbb{C},D_1)$ has chromatic number at least c, so by the de Bruijn–Erdős theorem [8] some finite graph G drawn in the plane with edge lengths in D_1 has $\chi(G)\ge c$; translate it so that its n vertices lie in $B(0,R_0)$, and put $R=2R_0$. If $\chi_c(W)=1$ there is nothing to prove. Otherwise let K be an integer with $1\le K<\chi_c(W)$ (any such K works, so a lower bound for $\chi_c(W)$ suffices), put $\varepsilon=1/(4nK)$, and let $N>0$ be a real algebraic number with

$$K\,|E(W)|\,\varepsilon^{-2}\sqrt{R/(\pi N\min D_2)}\;<\;1,$$

which holds for all large N. We show that $\chi(\mathbb{R}^2,D_1\cup ND_2)>K$. Suppose instead that $G(\mathbb{C},D_1\cup ND_2)$ has a proper k-colouring with $k\le K$. By Theorem 11 and step 6 of its proof, there are pairwise disjoint measurable sets $A_1,\dots,A_k$, each avoiding $D_1\cup ND_2$, whose union is conull. Let $\rho_i(x)=|A_i\cap B(x,R)|/(\pi R^2)$, a continuous function, and $S_i=\{x:\rho_i(x)\ge\varepsilon\}$.

By Lemma 14, since $1/(4nk) \geq \varepsilon$, every point lies in at least c of the sets S_i. Since $1_{S_i} \leq \rho_i/\varepsilon$ and A_i avoids Nd for $d \in D_2$, Lemma 15 gives $P_{Nd}(S_i) \leq \varepsilon^{-2} P_{Nd}(\rho_i) \leq \varepsilon^{-2}\sqrt{R/(\pi N d)}$. As $k \leq K < \chi_c(W)$, Lemma 16 gives

$$1 \;\leq\; k\,|E(W)|\,\varepsilon^{-2}\sqrt{R/(\pi N \min D_2)} \;<\; 1,$$

a contradiction. Taking $K = \chi_c(W) - 1$ gives the first inequality, and $\chi_c(W) \geq c\,\chi_{\mathrm{f}}(W)$ holds for every finite graph.

For the second statement, $c = \chi(\mathbb{R}^2, D_1)$ is finite (a product of hexagonal colourings, one for each distance, uses $7^{|D_1|}$ colours), and so is $\chi_c(\mathbb{R}^2, D_2) \leq c\,7^{|D_2|}$ (replace each colour by c new ones). A graph has a c-fold colouring with k colours if and only if it has a homomorphism into the finite Kneser graph of c-subsets of $\{1, \ldots, k\}$, so by compactness, as in the de Bruijn–Erdős theorem, some finite graph W drawn with edge lengths in D_2 has $\chi_c(W) = \chi_c(\mathbb{R}^2, D_2)$; apply the first statement to it. Finally $\chi_c(\mathbb{R}^2, D_2) \geq c \sup_W \chi_{\mathrm{f}}(W) = c\,\chi_{\mathrm{f},0}(\mathbb{R}^2, D_2) \geq c/m_{D_2}(\mathbb{R}^2)$ by Theorem 12. □

Proof of Corollary 9. By Corollary 2, $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2) > 1/0.2469975 > 4.0486$, so some finite unit-distance graph W has $\chi_{\mathrm{f}}(W) > 4.0486$. (1) Theorem 8 with $D_1 = D_2 = \{1\}$, $c = 6$ [22] and this W gives $\chi(\mathbb{R}^2, \{1, N\}) \geq \chi_6(W) \geq 6\,\chi_{\mathrm{f}}(W) > 24.29$ for large N. For every $N > 0$, $m_{\{1,N\}}(\mathbb{R}^2) \geq m_1(\mathbb{R}^2)^2$ (Bukh [2]; intersect a periodic set avoiding 1 with a suitable translate of a dilate by N of another), and the proof of the upper bound in Theorem 1 does not use algebraicity, so $\chi_{\mathrm{f}}(\mathbb{R}^2, \{1, N\}) \leq m_1(\mathbb{R}^2)^{-2} \leq 0.229364^{-2} < 19.01$. (2) Let $c_1 = 6$ and $D^{(1)} = \{1\}$. Given a finite set $D^{(j)}$ of integers with $\chi(\mathbb{R}^2, D^{(j)}) \geq c_j$, Theorem 8 with $D_1 = D^{(j)}$, $D_2 = \{1\}$ and this W gives an integer $d_{j+1} > \max D^{(j)}$ with $\chi(\mathbb{R}^2, D^{(j)} \cup \{d_{j+1}\}) > 4.0486\,c_j$; let c_{j+1} be the least integer above $4.0486\,c_j$. Then $c_2 = 25$, $c_3 = 102$, $c_4 = 413$ and $c_r \geq 6 \cdot 4.0486^{r-1}$, which exceeds $1.48 \cdot 4.0486^r$. Restricting colourings of $\mathbb{R}^d$ to a plane gives the bounds for $d \geq 2$. □

Remark (Explicit pairs). The proof is effective when G and W are explicit. Take $c = 5$, and for G de Grey's unit-distance graph with 1581 vertices, 7877 edges and no proper 4-colouring [9], whose vertices lie within distance 2.27 of one of them (we computed 2.2641 from exact coordinates). For W take the Moser spindle, with 7 vertices, 11 edges and independence number 2, so that $\chi_5(W) \geq 5 \cdot 7/2$, that is $\chi_5(W) \geq 18$. With $K = 17$, $n = 1581$, $\varepsilon = 1/(4nK) = 1/107508$ and $R = 4.54$, the condition $K\,|E(W)|\,\varepsilon^{-2}\sqrt{R/(\pi N)} < 1$ of the proof holds as soon as $N > (17 \cdot 11 \cdot 107508^2)^2 \cdot 4.54/\pi \approx 6.75 \cdot 10^{24}$. Hence $\chi(\mathbb{R}^2, \{1, N\}) \geq 18$ for every real algebraic $N \geq 10^{25}$; the best certified linear-programming bound for a pair is 12 (Table 1). For W one can also take the 607-vertex unit-distance graph of Bellitto, Pêcher and Sédillot, whose weighting has weighted independence ratio 512933/1999983 [3, §4.4], so that $\chi_{\mathrm{f}}(W) \geq 1999983/512933 > 3.8991$ (cf. [3, Thm. 2]) and $\chi_5(W) \geq 20$; only $\chi_{\mathrm{f}}(W) > 3.8$ is needed. With $K = 19$, $\varepsilon = 1/120156$ and the trivial bound $|E(W)| \leq \binom{607}{2} = 183921$, the condition holds for $N > 3.68 \cdot 10^{33}$, so $\chi(\mathbb{R}^2, \{1, N\}) \geq 20$ for every real algebraic $N \geq 10^{34}$, on the strength of their computation. These bounds avoid OpenAI's theorem $\chi(\mathbb{R}^2) \geq 6$ but still rest on Theorem 11, that is, on OpenAI's rigidity theorem.

6. Linear-programming certificates

Proposition 17 (Oliveira Filho–Vallentin [21], with clique inequalities)**.** *Let $D = \{d_1, \ldots, d_\ell\}$ and let $W_1, \ldots, W_r$ be finite planar point sets, each with all pairwise distances in D, and $p_1, \ldots, p_r$ points. Suppose $z_0, \ldots, z_\ell \in \mathbb{R}$ and $y_1, \ldots, y_r \geq 0$ satisfy*

$$g(t) = z_0 + \sum_k z_k J_0(d_k t) + \sum_c y_c \sum_{w \in W_c} J_0(|w - p_c|\,t) \;\geq\; 0 \qquad (t \geq 0),$$

with $s = z_0 + \sum_k z_k + \sum_c y_c |W_c| > 0$. Then $m_D(\mathbb{R}^2) \leq (z_0 + \sum_c y_c)/s$.

Proof. By Lemma 13 it suffices to consider a periodic measurable set A of density $\delta > 0$ avoiding D. Its autocorrelation $\varphi(v) = \delta(A \cap (A - v))$ is positive definite, with spectral measure ν on the dual lattice, $\nu(\{0\}) = \delta^2$, total mass δ, so that $\varphi(v) = \int e^{2\pi i \langle \xi, v \rangle}\, d\nu(\xi)$; its radial average is $f(r) = \int J_0(2\pi r|\xi|)\, d\nu(\xi)$, with $f(0) = \delta$ and $f(d_k) = 0$. For a set W with pairwise distances

in D and any p, the sets $\{x \in A : x + w - p \in A\}$, $w \in W$, are disjoint subsets of A, so $\sum_w \varphi(w-p) \le \delta$; averaging over rotations of W about p gives $\sum_w f(|w-p|) \le \delta$. Integrating $g(2\pi|\xi|) \ge 0$ against ν, $\delta^2 g(0) \le \int g(2\pi|\xi|)\,d\nu = z_0\delta + \sum_c y_c \sum_w f(|w-p_c|) \le (z_0 + \sum_c y_c)\delta$, and $g(0) = s$. Without clique terms this is [21, Thm. 1.1]; the clique terms are in the spirit of [21, §3.1], which uses simplices, while we also use a rhombus. □

We solve the linear program in floating point, round the coefficients to rationals and verify $g \ge 0$ rigorously. Three programs check every certificate.

- `certify_lp2.py` uses Arb ball arithmetic, with distances and radii enclosed from their exact expressions. On $[0, T]$ it walks with adaptive steps: at a point t it encloses $g(t)$ and moves on by about $g(t)/L$, where $L = \sum |\text{coefficient}| \cdot \text{radius}/\sqrt{2}$ bounds $|g'|$ because $|J_1(x)| \le 1/\sqrt{2}$ for real x [11, 10.14.1]. On $[T, \infty)$ it uses $|J_0(x)| \le \sqrt{2/(\pi x)}$ for $x > 0$: $u = \sqrt{x}J_0(x)$ solves $u'' + (1 + 1/(4x^2))u = 0$, so $u^2 + u'^2/(1 + 1/(4x^2))$ is nondecreasing (Sonin–Pólya) with limit $2/\pi$. The clique distances are checked exactly with a computer algebra system, and the bound is an exact fraction.
- `check2.py`, written separately, repeats the check with 60-digit floating point and its own sweep.
- `check3.py`, adapted from the checker of one of the referees, encloses g by Arb over whole intervals of t, bisecting until each enclosure is nonnegative, and builds the cliques and their radii in exact arithmetic of $\mathbb{Q}(\sqrt{3})$.

D	clique sets	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1, 2\}$	$6 + 2$ triangles	$0.159222 < 1/6$	$[0, 1297]$	$\chi \ge 7$
$\{1, \sqrt{3}\}$	triangle, rhombus	$0.162866 < 1/6$	$[0, 757]$	$\chi \ge 7$
$\{1, (\sqrt{6}+\sqrt{2})/2\}$	none	$0.166018 < 1/6$	$[0, 107]$	$\chi \ge 7$
$\{1, 10\}$	none	$0.149599 < 1/6$	$[0, 36]$	$\chi \ge 7$
$\{1, 11/6\}$	none	$0.140995 < 1/7$	$[0, 171]$	$\chi \ge 8$
$\{1, 46/25\}$	none	$0.139185 < 1/7$	$[0, 175]$	$\chi \ge 8$
$\{1, 40\}$	none	$0.115511 < 1/8$	$[0, 36]$	$\chi \ge 9$
$\{1, 100\}$	none	$0.103302 < 1/9$	$[0, 37]$	$\chi \ge 10$
$\{1, 1000\}$	none	$0.089085 < 1/11$	$[0, 37]$	$\chi \ge 12$

TABLE 1. Certified density bounds (rounded up). For $\{1, 2\}$ the clique sets are six unit equilateral triangles and two equilateral triangles of side 2, each at one position; for $\{1, \sqrt{3}\}$, one unit equilateral triangle and one rhombus made of two such triangles (diagonals 1 and $\sqrt{3}$). All three programs accept every certificate.

D	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1, 40, 1600\}$	$0.046389 < 1/21$	$[0, 36]$	$\chi \ge 22$
$\{1, 10, 100, 1000\}$	$0.039641 < 1/25$	$[0, 37]$	$\chi \ge 26$
$\{1, 100, 10^4\}$	$0.037168 < 1/26$	$[0, 37]$	$\chi \ge 27$
$\{1, 30, 900, 27000\}$	$0.021264 < 1/47$	$[0, 36]$	$\chi \ge 48$

TABLE 2. Certified density bounds for three and four distances (rounded up), without clique terms. `certify_lp2.py` and `check3.py` accept every certificate; the sweeps have up to $2.6 \cdot 10^7$ steps.

For $\{1, \sqrt{3}\}$ Oliveira Filho and Vallentin [21, §4] found the bound 0.170213 without clique terms, and 0.141577 for $\{1, j_{1,2}/j_{1,1}\}$, where $j_{1,k}$ are zeros of J_1; that ratio is presumably transcendental, so Theorem 11 does not apply to it directly, but nearby rationals such as 11/6 and 46/25 do. For the golden-ratio pair $\{1, (1+\sqrt{5})/2\}$ of [23], the program with regular unit pentagons (whose sides and diagonals are 1 and $(1+\sqrt{5})/2$) and triangles as cliques gives only 0.1769, above 1/6.

7. Uniform certificates for two distances

For $d \geq 1$ and real numbers z_0, z_1, z_2 put $g_d(t) = z_0 + z_1 J_0(t) + z_2 J_0(dt)$ and $s = z_0 + z_1 + z_2$. By Proposition 17 without clique terms, if $s > 0$ and $g_d \geq 0$ on $[0, \infty)$, then $m_{\{1,d\}}(\mathbb{R}^2) \leq z_0/s$. Besides $|J_0(x)| \leq \sqrt{2/(\pi x)}$ for $x > 0$ (Section 6) we use $|J_0'| = |J_1| \leq 1/\sqrt{2}$ and $|J_0''| = |J_0 - J_2|/2 \leq 1$ on $\mathbb{R}$ [11, 10.14.1], and $J_0 \geq -\mu$ on $[0, \infty)$ with $\mu = 0.4028$; the minimum of J_0 is $J_0(j_{1,1}) = -0.402759\ldots$, and the programs check $J_0 \geq -\mu$ in interval arithmetic on $[0, 4]$, while beyond 4 it follows from $\sqrt{2/(4\pi)} < 0.399$.

Lemma 18. *Let $D > 0$, $z_2 \geq 0$ and $s > 0$, and put $e(y) = \min\{\mu, \sqrt{2/(\pi y)}\}$ for $y > 0$. If*

$$h(t) = z_0 + z_1 J_0(t) - z_2\, e(Dt) \;\geq\; 0 \qquad (t > 0),$$

then $g_d \geq 0$ on $[0, \infty)$ for every $d \geq D$, so that $m_{\{1,d\}}(\mathbb{R}^2) \leq z_0/s$ for every $d \geq D$.

Proof. For $x \geq y > 0$ we have $J_0(x) \geq -\mu$ and $J_0(x) \geq -\sqrt{2/(\pi x)} \geq -\sqrt{2/(\pi y)}$, so $J_0(x) \geq -e(y)$. For $d \geq D$ and $t > 0$ apply this with $x = dt$ and $y = Dt$; since $z_2 \geq 0$, $g_d(t) \geq h(t) \geq 0$. Finally $g_d(0) = s > 0$. □

So one function serves every $d \geq D$, and checking it is a problem in one variable, in which nothing oscillates faster than $J_0(t)$. Table 3 lists the seven certificates. Each is checked on $[0, 10]$, and on $[10, \infty)$ by $h(t) \geq z_0 - |z_1|\sqrt{2/(10\pi)} - z_2\sqrt{2/(10\pi D)} > 0$.

k	D	certified bound z_0/s for all $d \geq D$	boxes
7	6.52	$0.1666308 < 1/6$	297
8	12.3	$0.1428093 < 1/7$	169
9	24.3	$0.1249785 < 1/8$	161
10	52.9	$0.1111073 < 1/9$	237
11	140.2	$0.0999955 < 1/10$	139
12	601	$0.0909057 < 1/11$	97
13	54 300	$0.0833328 < 1/12$	75

Table 3. The certificates of Lemma 18 (bounds rounded up; `tail_k`k`.json`), each giving $\chi(\mathbb{R}^2, \{1, d\}) \geq k$ for every real algebraic $d \geq D$. The last column is the number of boxes of `tail.py` on $[0, 10]$.

Remark. These thresholds are near the limit of the method. Put $\mu_0 = -\min J_0 = 0.40275\ldots$, let $d \geq 20$, and normalise $s = 1$. If $g_d \geq 0$, its values at $t = j_{1,1}/d$ and at a point t with $|t - j_{1,1}| \leq \pi/(2d)$ and $J_0(dt) = 0$ (there is one, because $dj_{1,1} > j_{0,1}$ and consecutive zeros of J_0 are less than π apart) give

$$z_0 + a z_1 - \mu_0 z_2 \geq 0 \qquad \text{and} \qquad z_0 + b z_1 \geq 0,$$

with $a = J_0(j_{1,1}/d) \in (0, 1]$ and $|b + \mu_0| \leq \eta = \pi^2/(8d^2) < \mu_0$, because $J_0'(j_{1,1}) = 0$ and $|J_0''| \leq 1$. If $z_1 < 0$ the first inequality gives $z_0 > \mu_0/(1 + \mu_0)$; otherwise the two give

$$z_0 \;\geq\; \frac{\mu_0(\mu_0 - \eta)}{a + \mu_0 + (1 + \mu_0)(\mu_0 - \eta)} \;>\; \frac{1}{12.2}.$$

As $d \to \infty$ the right-hand side tends to $L = \mu_0^2/(1 + \mu_0)^2 = 1/12.13\ldots$, and L is the limit of the program: with μ_0 in place of μ in Lemma 18, the coefficients $z_1 = \mu_0/(1 + \mu_0)^2$, $z_2 = 1/(1 + \mu_0)$ and $z_0 = L + \delta$, where $\delta = z_2\sqrt{2/(\pi D)} + z_1(y_0/D)^2$ and $y_0 = 2/(\pi\mu_0^2)$, satisfy $h \geq 0$ and give the bound $(L + \delta)/(1 + \delta)$ for every $d \geq D$. Indeed $L = \mu_0 z_1$ and $\mu_0 z_2 = (1 + \mu_0)z_1$, so $h(t) = \delta + z_1(J_0(t) - 1)$ for $Dt \leq y_0$, $h(t) = \delta + z_1\big(\mu_0 + J_0(t) - (1 + \mu_0)\sqrt{y_0/(Dt)}\big)$ for $Dt \geq y_0$, and $J_0(t) \geq 1 - t^2/4$ for $t \leq 2$ and $\mu_0 + J_0 \geq 0$ give $h \geq 0$ (separately for $Dt \leq y_0$, for $y_0 \leq Dt \leq 2y_0$, for $2y_0 \leq Dt$ with $t \leq 1$, and for $t \geq 1$). Hence for $d \geq 20$ the program never gives more than 13 colours, and in floating point it never does for any d.

For a bounded range of d we certify $g_d \geq 0$ for all d in an interval $[a, b]$ at once. On a box $[a, b] \times [t_0, t_1]$ the argument dt lies in $[at_0, bt_1]$, and J_0 on an interval $[m - r, m + r]$ is enclosed by $J_0(m) \pm (|J_1(m)|r + r^2/2)$, by $[-\mu, 1]$, and by $\pm\sqrt{2/(\pi x_0)}$ if the interval starts at $x_0 > 0$. The program bisects $[0, T]$ in t until every box has a nonnegative lower bound for g_d, and on $[T, \infty)$ it uses $g_d(t) \geq z_0 - |z_1|\sqrt{2/(\pi T)} - |z_2|\sqrt{2/(\pi a T)} > 0$. Three files of such certificates cover $[1.75, 1.93]$ with $k = 7$ (19 intervals), $[1.83, 1.855]$ with $k = 8$ (5 intervals) and $[12.05, 12.3]$ with $k = 8$ (11 intervals), which completes the proof of Theorem 5.

`tail.py` and `uniform.py` choose the coefficients by a floating-point linear program, round them to fractions, raise z_0 by a third of the slack, and prove $h \geq 0$, respectively $g_d \geq 0$ on each interval, with the enclosures above in Arb ball arithmetic; the same programs re-verify stored certificates, and `tail.py` reproduces its certificates byte for byte. `check_uniform.py`, written separately, recomputes every target $1/(k-1)$ from k, checks that the intervals of a file are contiguous and cover its range, encloses J_0 by point values and the Lipschitz bound $1/\sqrt{2}$ instead of the Taylor bound, bisects boxes in both t and d, sweeps h with steps chosen from its value, and checks $J_0 \geq -\mu$ itself. Both accept all ten certificates, and both reject corrupted ones: a lowered z_0, a scaled z_2, a threshold D moved below the certified one, a raised k, a gap between intervals, a reversed interval, an interval extended below its range, and a certificate whose g_d has minimum -10^{-5} at a point that a sampling scan misses.

8. Remarks

(1) The hard input is OpenAI's rigidity theorem [22, Thm. 2.3], which OpenAI formalized in Lean. Our contribution is the observation that their transfer holds for every set of algebraic distances and for fractional colourings, the consequences of combining it with density bounds, and the two-scale argument of Section 5, which uses OpenAI's theorem a second time, as a statement about finite graphs.

(2) Davies [7] proves, by a different method, that in $\mathbb{R}^d$ with $d \geq 4$, and for finite D whose elements have rational squares, the chromatic number equals the measurable chromatic number and the independence ratio equals the measurable one; his certificates come from the completely positive framework of DeCorte, Oliveira Filho and Vallentin and are transferred through lattice embeddings. With Bukh's theorem this gives exponential growth in k for $d \geq 4$, and he hopes to extend the method to $d = 2, 3$. Our results concern $d = 2$ (and every $d \geq 2$ for Corollaries 6 and 9, by restriction to a plane), allow arbitrary algebraic distances, and rest on OpenAI's transfer.

(3) The distances must be algebraic because [22, Thm. 2.3] is a statement about the algebraic plane. This is a limitation of the method, but probably also of the statement: if Bukh's conjecture above holds, the inequality $\chi(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$ fails for some algebraically independent D.

(4) No explicit graph is given. The graph of Dúcz and Varga can in principle be made explicit but is astronomically large; an explicit unit-distance graph of reasonable size with fractional chromatic number above 4, or an explicit graph with edges of lengths 1 and 2 and no proper 6-colouring, remains to be found.

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